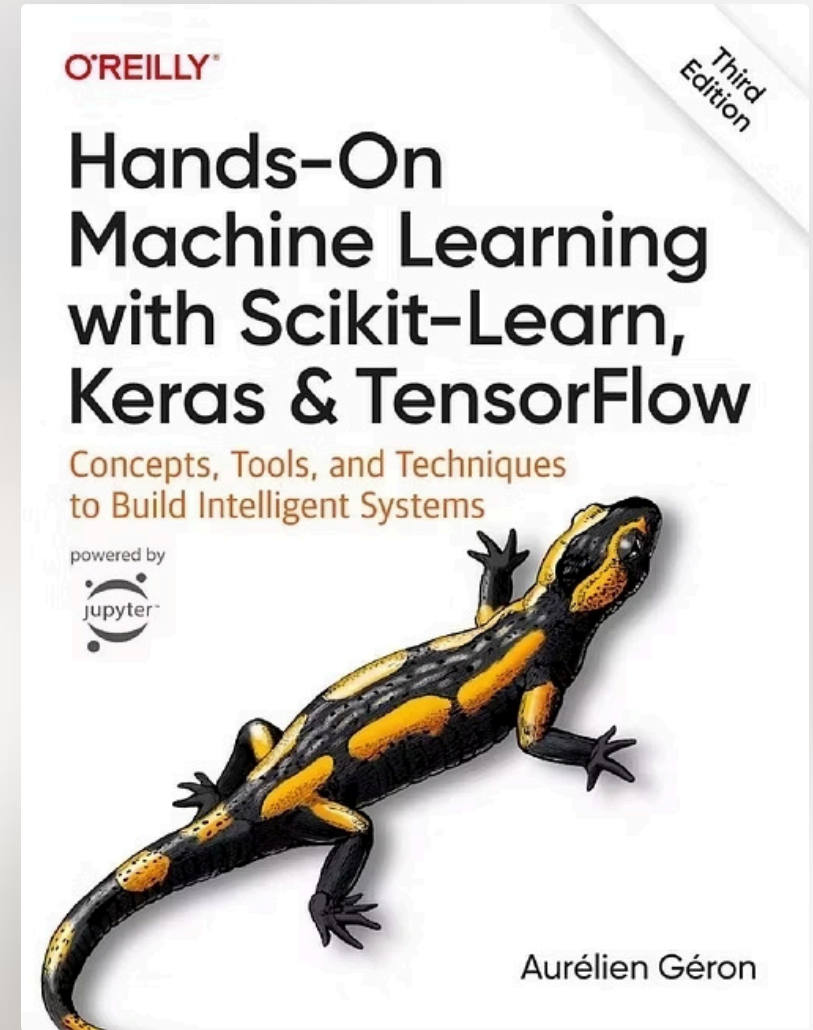


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

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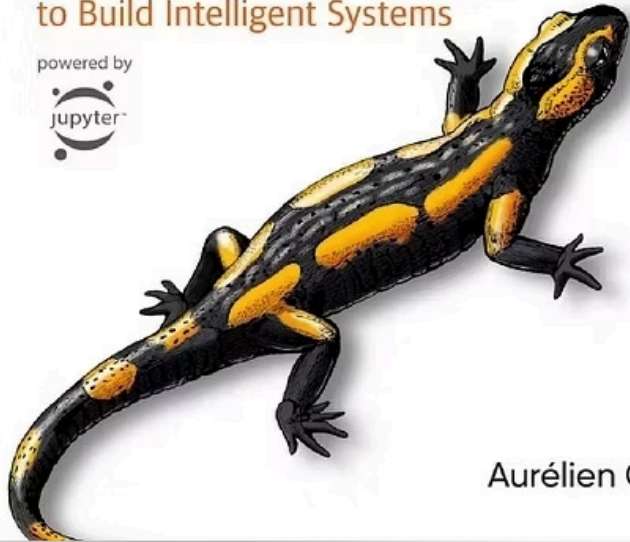
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Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
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Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. **Main Challenges of Machine Learning**
6. Testing and Validating



A futuristic robot with a glowing blue eye and a tablet displaying code.

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - **Overfitting the Training Data**
 - Underfitting the Training Data
 - Stepping Back: Reviewing the Chapter So Far
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch

Main Challenges of Machine Learning

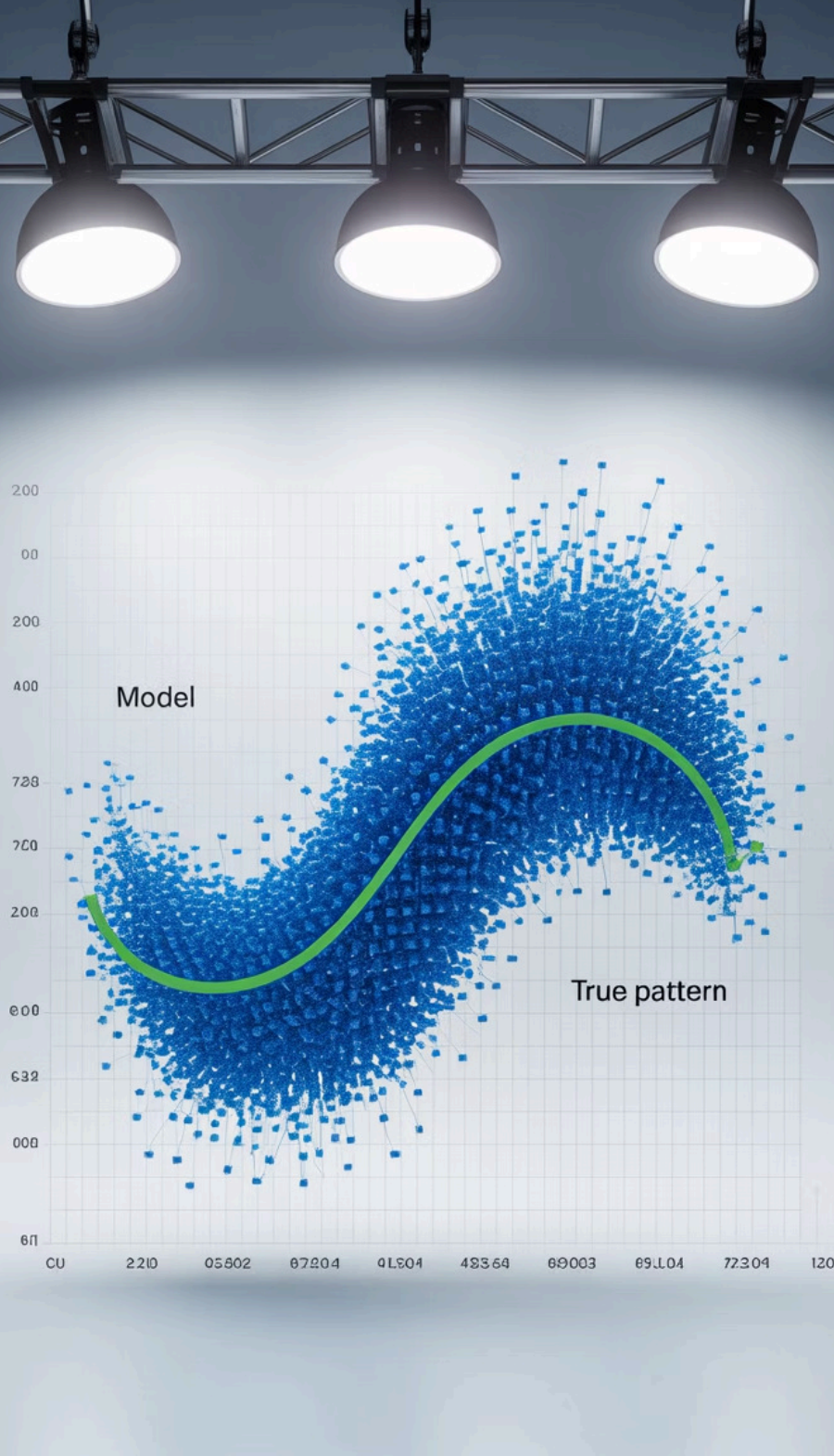
Insufficient Quantity of Training Data, Non-representative Training Data, Poor-Quality Data, Irrelevant Features, **Overfitting the Training Data**, Underfitting the Training Data, Stepping Back: Reviewing the Chapter So Far

Overfitting the Training Data: Overview

What is Overfitting?

Overfitting occurs when a **model performs well on the training data** but **does not generalize well to new, unseen data**.

Example: A taxi driver ripping you off may lead to the **overgeneralization** that **all taxi drivers in that country are thieves**.



Example of Overfitting: Polynomial Model

High-Degree Polynomial Model:

A complex model (like a **high-degree polynomial**) can **overfit** the training data by **closely following all data points**.

Result: Although the **model** may **perform well** on the **training data**, its **predictions on new data** may not be **reliable**.

Visual Example

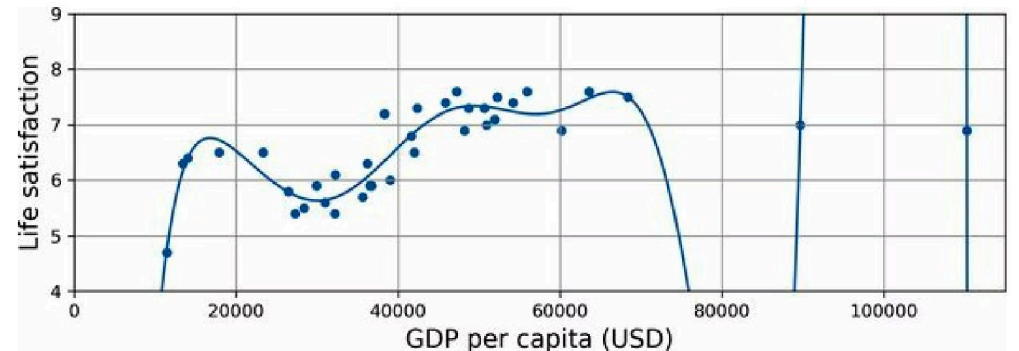


Figure 1-23: A high-degree polynomial model that overfits the training data (shown by the solid line).

Why Does Overfitting Occur?

Subtle Patterns in Noisy Data:

- Complex models like **deep neural networks** can detect very subtle patterns
 - But if the **training set is noisy or small**, the model may "**learn**" **patterns** that are **irrelevant**.

Example of Overfitting with Irrelevant Features:

Adding irrelevant features (like **country names**) could cause the model to detect **meaningless patterns** (e.g., countries with a "w" in their name have **high life satisfaction**).

Pattern Example: New Zealand, Norway, Sweden, and Switzerland have high satisfaction, but this pattern will likely **not generalize** to countries like **Rwanda** or **Zimbabwe**.

Regularization: Reducing Overfitting

What is Regularization?

- Regularization adds a **constraint** to the model to **reduce its complexity** and **prevent overfitting**.
- The **regularized model** has a **smaller slope**, leading to **better generalization**.



3 Models for the Life Satisfaction Training Data

1. The **dotted line** represents the **original model** that was trained on the **countries** represented as **circles** (without the countries represented as **squares**)
2. The **solid line** is the model trained with **all countries** (circles and squares), and
3. The **dashed line** is a model trained with the **same data as the first model** but with a **regularization constraint**.

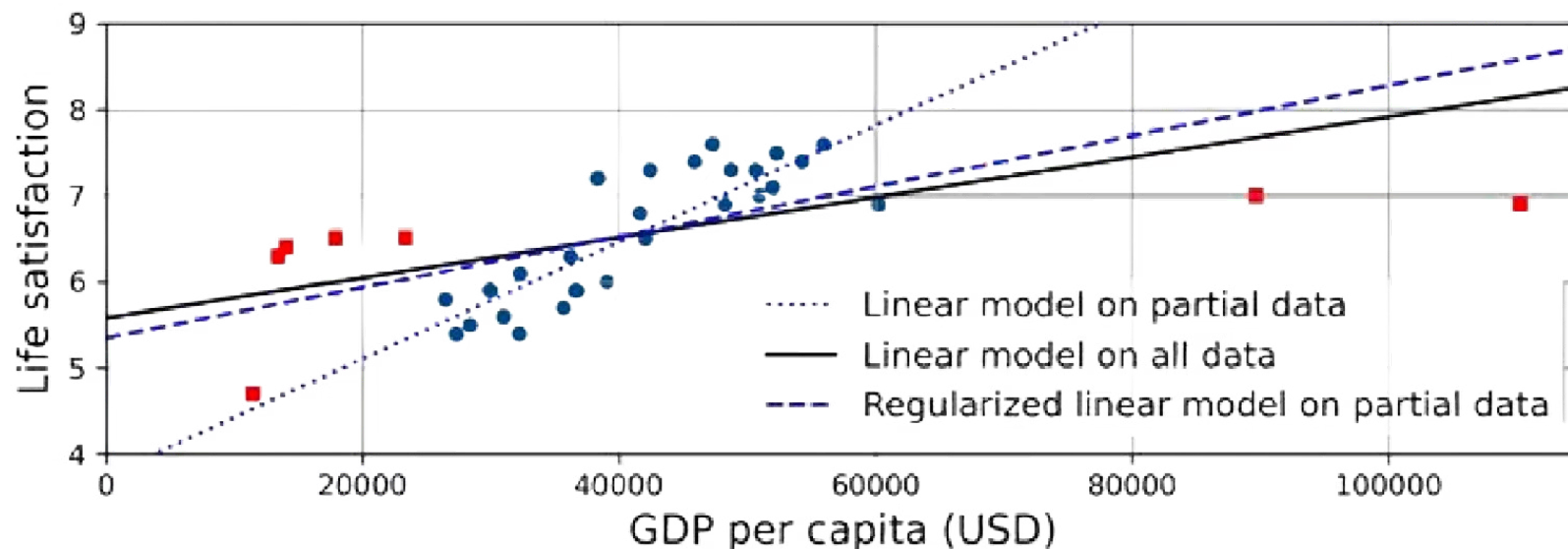


Figure 1-24: Regularization forces the model to fit the data more simply, improving its ability to generalize to new data (squares).

Controlling Regularization: Hyperparameters

- The amount of regularization to apply during learning can be controlled by a hyperparameter.



Hyperparameters

- Parameters of the learning algorithm (not of the model).
- Hyperparameter Setup:
 - Must be set before training and remain constant during training.



Effect of Hyperparameter Tuning

- **Large Regularization Value:** The model becomes simpler (almost flat - slope close to zero), preventing overfitting but may miss a good solution.
- **Tuning Hyperparameters:** Proper tuning is essential to finding a balance between model complexity and accuracy.

Key Takeaways

Overfitting

Overfitting occurs when the **model fits the training data too closely** and **fails to generalize**.

Hyperparameter Tuning

The **amount of regularization** can be **controlled by adjusting the regularization hyperparameter** before **training**.



Right Balance

You want to find the right balance between **fitting the training data perfectly** and **keeping the model simple enough to ensure that it will generalize well**.

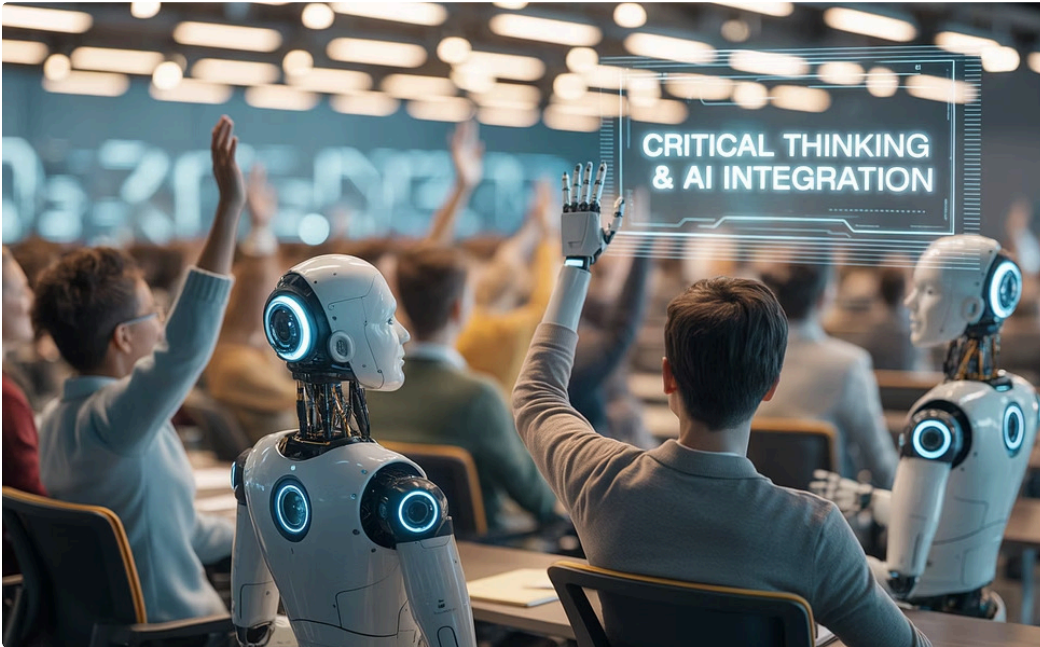
Regularization

Regularization helps **reduce overfitting** by **adding a constraint to simplify the model**.



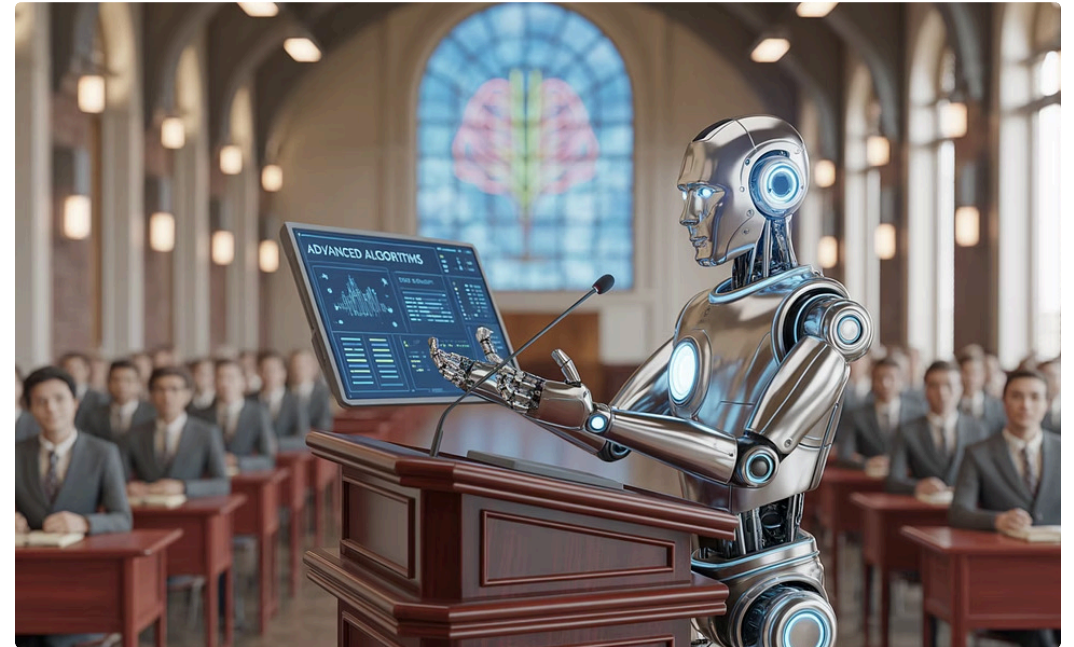
Next: Underfitting the Training Data

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts